Pedro Luiz R. Vaz de Melo

FOREST ENGINEER | GEOSPATIAL DATA SCIENTIST | GIS & REMOTE SENSING | PYTHON | GOOGLE EARTH ENGINE | POSTGIS | REST APIS

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LinkedIn GitHub Portfolio

SUMMARY

Forest Engineer (UFSJ, 2025) and Geospatial Data Scientist with a solid foundation in back-end development. I build pipelines that turn raw environmental data (satellite imagery and official Brazilian datasets such as IBGE, SICAR and MapBiomas) into decision products: nationwide H3 grids, WebMaps, dashboards and APIs. At EnvironBIT, I maintain environmental feature pipelines for about 10.8 million hexagons covering Brazil; as a consultant for power transmission line projects, I automate vegetation-suppression inventories, licensing cartography and terrain analyses. My workflows cut technical report production from days to minutes, with tested, reproducible code.

OBJECTIVE

Seeking roles as Geospatial Data Analyst, Data Scientist, GIS/Geoprocessing Specialist or GIS Developer, applying geospatial Python, Google Earth Engine, PostGIS and machine learning to challenges in forestry, agri-environmental, infrastructure and sustainability sectors.

EXPERIENCE

Data Scientist | EnvironBIT (CNPq Project)
Apr 2026 – Present · Remote
Research and development in environmental data intelligence for road infrastructure and wildlife conservation.

- Develop and maintain environmental feature pipelines on an H3 hexagonal grid (resolution 8, about 10.8 million cells across Brazil), combining raster features (CHELSA and CGIAR climate, land use and land cover, ecoClimate, PRODES) and vector features (IBGE, ANA, SGB): more than 230 features per hexagon, with automated tests and Medallion-architecture storage (MinIO and PostgreSQL). These datasets feed the team's wildlife-vehicle collision and road accident models.
- Built and keep evolving the Integrated Data Panel (Painel Integrado de Dados), an internal dashboard running on Node.js, TypeScript, PostgreSQL and Docker: a nationwide view of about 32 thousand hexagons, drill-down with typical latency of 200 ms, LDAP/JWT authentication, caching and CSV export.
- Produce road-level wildlife-vehicle collision (WVC) risk extracts, such as about 8,200 km of Minas Gerais state highways, intersecting the model's H3 cells with each road segment and with IBGE urbanization levels.

Geoprocessing & GIS Development Consultant | Vimef Engenharia
Feb 2026 – Present · Remote

- Operational WebMaps for vegetation-suppression inventories on power transmission lines: on a 203 kV line, 74 ha of eucalyptus and pine mapped across 44 spans, with in-field polygon editing and permanent preservation areas (APP) excluded from totals; on a 500 kV line (about 323 km and 599 towers), a pre-suppression planning map with Sentinel-2 and MapBiomas.
- Licensing cartography following the Minas Gerais State Forestry Institute (IEF) terms of reference for compensation and degraded-area recovery plans (PCIA and PRADA) on a 58.4 km, 345 kV line, with map sheets generated automatically in Python from MapBiomas, ANA (BHO), FABDEM and IDE-SISEMA.
- Slope analyses using EMBRAPA classes and true 3D surface area from LiDAR, drone photogrammetry and elevation models (ALOS PALSAR 12.5 m and Copernicus 30 m).
- Digital Timber Tally (Romaneio Digital): an offline-first web system for log scaling (Smalian formula) and firewood in the field, with a client dashboard published on GitHub Pages.

Python (Back-end) and Data Science Intern | Instituto Espinhaço
Aug 2025 – Oct 2025 · Remote

- Architected a suite of Python ETL pipelines combining local processing (GeoPandas, Rasterio) with Google Earth Engine over governmental and territorial datasets (SICAR, IBGE, MapBiomas).
- Fully automated the generation of technical PDF dossiers (rural environmental registry, legal reserve, land use, slope and climate), cutting each report from days to minutes.

EDUCATION

Jun 2026 – Present
Postgraduate Degree in Data Science, Data Science Academy in partnership with Facint College

Mar 2019 – Dec 2025
B.Sc. in Forestry Engineering, Universidade Federal de São João del-Rei (UFSJ)

COURSES & CERTIFICATIONS

- Python Language Fundamentals – From Basics to AI Applications (96h) | Data Science Academy
- Microsoft Power BI for Business Intelligence and Data Science (72h) | Data Science Academy
- Developing Interactive Web GIS Platforms with Server Side Technologies (18h) | Udemy
- Integrating QGIS with PostgreSQL/PostGIS (11.5h) | Udemy
- Getting Started with PythonGIS (10.5h) | Udemy
- Increasing Productivity with PyQGIS (8.5h) | Udemy
- Create REST APIs with Python and Django REST Framework (9.5h) | Geek University
- Journey to the Future with Node.js Bootcamp (40h) | DIO
- Klabin Bootcamp: Data Analysis with Excel and Power BI (90h) | DIO & Klabin
- Environmental Forensics (30h) | Instituto Brasileiro de Sustentabilidade (INBS)
- ESG: What Every Professional Should Know (8h) | INBS

- Environmental Licensing (8h) | INBS
- Carbon Markets and Credits (8h) | INBS
- **English Certification | EF SET (Score: 77/100, C2 Proficient)**
- **Invited speaker: keynote lecture "Cloud geospatial ETL with Google Earth Engine" | Future Tech Leaders Brazil Bootcamp (Brazilian Space Agency and SDG AI Lab), 2026**
- Teaching Assistant for Forest Genetics and Breeding | Aug 2023 – Dec 2023
- Teaching Assistant for Forest Pathology | Feb 2024 – Jul 2024

SKILLS

- **Languages & Web:** Python, SQL, JavaScript, TypeScript, R, HTML/CSS.
- **Geotechnology & Remote Sensing:** Google Earth Engine, QGIS (PyQGIS), PostGIS, GeoPandas, Rasterio, Shapely, GDAL, H3, GeoServer, WebGIS; Sentinel-2, MapBiomas, CHIRPS, ERA5-Land, LiDAR and elevation models.
- **Data Engineering & Backend:** ETL/ELT, Parquet, DuckDB, MinIO (Medallion architecture), FastAPI, Node.js (Express, Fastify), Django REST Framework, SQLModel/SQLAlchemy, Pydantic, RESTful APIs, Docker.
- **Data Science & AI:** Pandas, NumPy, SciPy, scikit-learn, PyTorch (MLP, CNN, U-Net), spatial statistics (Moran's I), LangChain, RAG (Chroma), LLM integration.
- **Databases:** PostgreSQL/PostGIS, DuckDB, MySQL, SQLite.
- **Visualization & BI:** Leaflet, Chart.js, Plotly, Streamlit, Power BI, Excel (advanced).
- **Quality & Operations:** Git/GitHub, GitHub Actions, pytest, OpenTelemetry, Prometheus, Grafana, Postman/Insomnia.
- **Domain Knowledge:** Environmental licensing (PCIA, PRADA, APP), CAR/SICAR (Brazilian Rural Environmental Registry), forest inventory, ESG, IBGE data (SIDRA).

LANGUAGES

Portuguese	**English**	**Spanish**
Native	Fluent (C2)	Intermediate

FEATURED PROJECTS

Public Forests of Brazil (WebGIS) | Python, GeoPandas, Shapely, Leaflet.js, Chart.js | Analysis of Brazil's National Public Forest Registry (CNFP 2024) combined with IBGE boundaries: 339.8 million hectares (39.91% of the country) and 20,829 features, published as a four-tab WebMap with state and municipal choropleths, a municipal ranking, a historical series since 1930 and a breakdown by biome. **Access:** pedroluizskt.github.io/florestas-publicas-brasil

ENSO in Brazil's Center-West: El Niño × climate teleconnection | Python, Google Earth Engine, GeoPandas, SciPy, Leaflet.js | Python package that measures the teleconnection between El Niño and the climate of 468 municipalities in MT, MS, GO and DF, combining NOAA's RONI, ONI and SOI indices with CHIRPS precipitation and ERA5-Land temperature. Applies Bretherton autocorrelation correction, permutation tests and Moran's I (above 0.93 for the latitudinal gradient), with a dashboard that works offline. **Access:** pedroluizskt.github.io/enso-centro-oeste

Extreme Rainfall Monitoring in Minas Gerais (WebGIS) | Python, Google Earth Engine, GeoPandas, Leaflet.js | Serverless pipeline that processes 20 years of daily CHIRPS data in Earth Engine and quantified the February 2026 climate disaster in Minas Gerais: Juiz de Fora recorded 558 mm, 183% above its historical mean. Used as the case study in the FTL Brazil keynote lecture. **Access:** pedroluizskt.github.io/gee-precipitation-webmap-mg

Open Buildings Explorer BR | Python, DuckDB, spatial SQL, Parquet (S3), Leaflet.js, pytest | Queries the 141 million buildings of the Google-Microsoft Open Buildings dataset in Brazil directly from S3, with predicate pushdown and spatial SQL, without downloading the dataset. The WebMap compares both sources by area of interest; 169 automated tests. **Access:** pedroluizskt.github.io/open-buildings-explorer-br

Forest-Py: Machine Learning in Forest Engineering | Python, PyTorch, scikit-learn, pytest | Installable package (forestpy) comparing classical models and neural networks (MLP, CNN, U-Net) for dendrometry, forest inventory and remote sensing across 12 sessions; the U-Net beats the NDVI baseline by +0.72 IoU in tree crown segmentation. 223 automated tests. **Access:** github.com/PedroLuizskt/forest-py-estudos

GeoPython Assistant (AI for GIS) | Python, Streamlit, LangChain, Llama-3 70B (Groq), Chroma, Pydantic v2 | Geospatial coding assistant. Version 1, published on Streamlit, uses prompt engineering on Llama-3 70B; version 2 combines a structured diagnosis of the user's files with RAG over the official GeoPandas, Rasterio and Rioxarray documentation to reduce hallucinated APIs. **Access:** geopython-assistant-skt.streamlit.app · github.com/PedroLuizskt/GeoPython-Assistant-v2

Data Science & Business Analytics Portfolio | Power BI, DAX, M Language, SQL, Python, R | Dimensional modeling (Star Schema), ETL pipelines and executive dashboards for finance, marketing and logistics, with algorithms embedded in the visuals: customer segmentation with K-Means (Python) and fraud detection with Isolation Forest (R). **Access:** github.com/PedroLuizskt/data-analytics-portfolio

Full portfolio, with live maps and consulting work: pedroluizskt.github.io